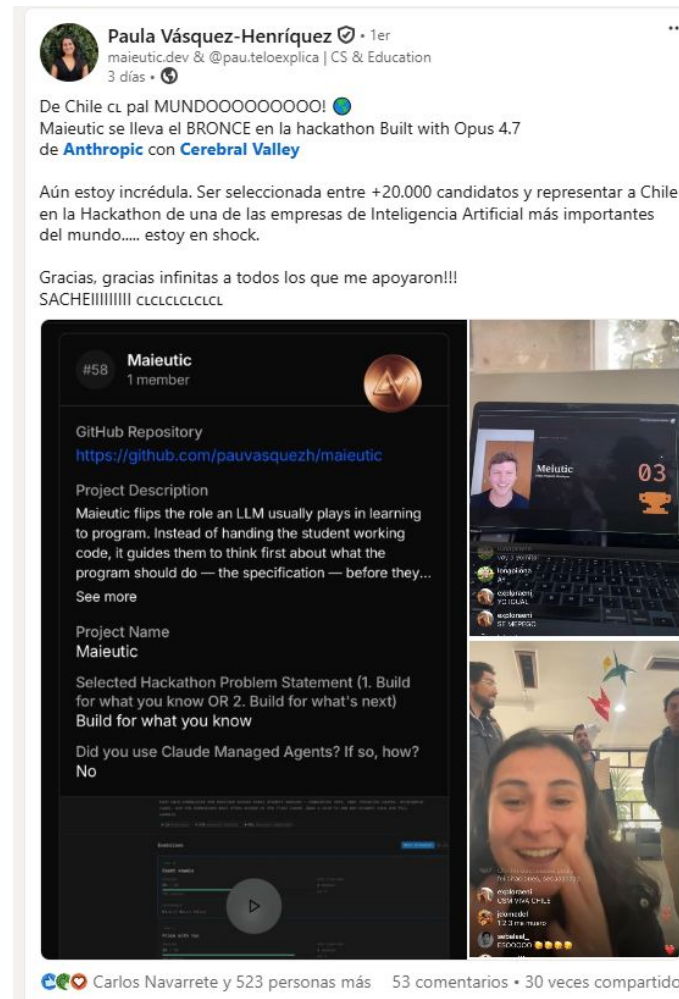


Ingeniería de soluciones con AI

Semestre 2026-2

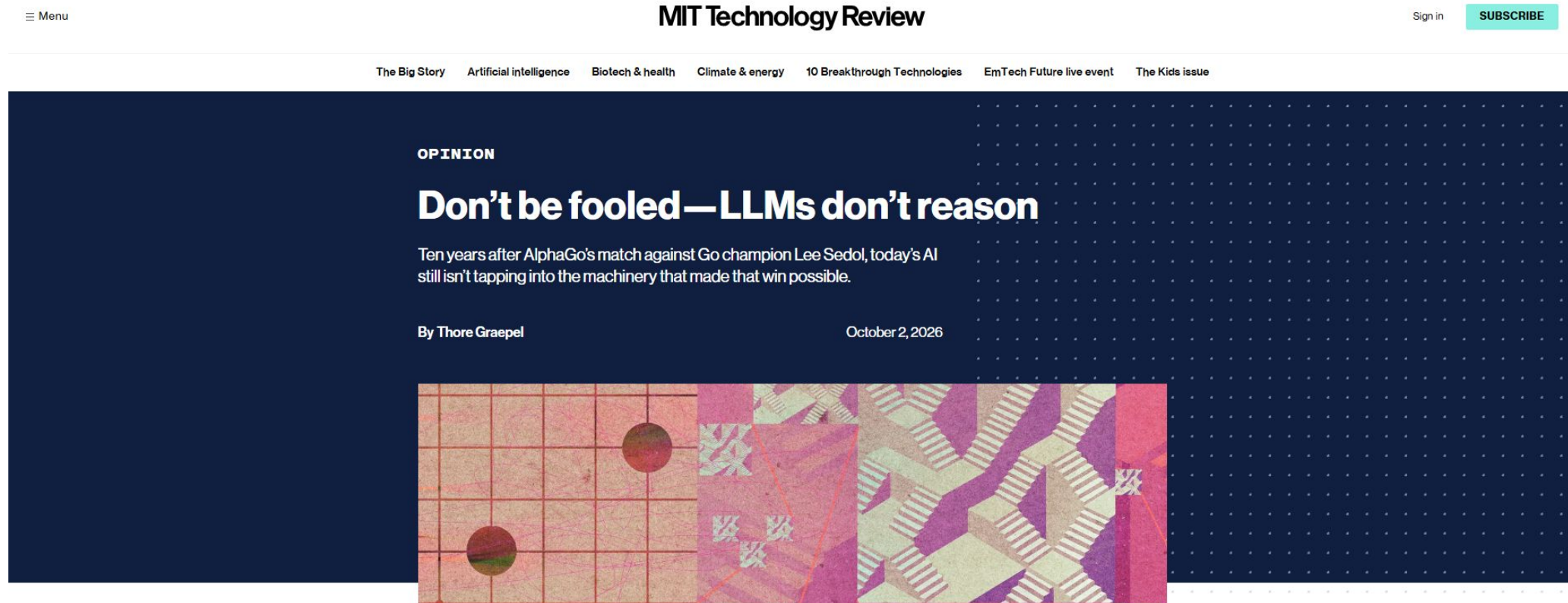
Docente:
Francisco Macaya

Noticias: Una chilena quedo tercera en la hackathon de Anthropic.



*Resolver problemas que importan,
y partan del problema, no de la
solución*

Noticias: Los LLMs no razonan, pero necesitamos trazabilidad para temas importantes como humanos.



Fuente: <https://www.technologyreview.com/2026/10/02/1145639/dont-be-fooled-llms-dont-reason/>

Noticias: Summit Fundación Pais Digital



- *Involucrar su entorno y la educación para resolver problemas que a ellos les interesa.*
- *Alta expectativas en los estudiantes*
- *Mantener las expectativas grandes.*

¿Cómo evaluar nuestros sistemas?

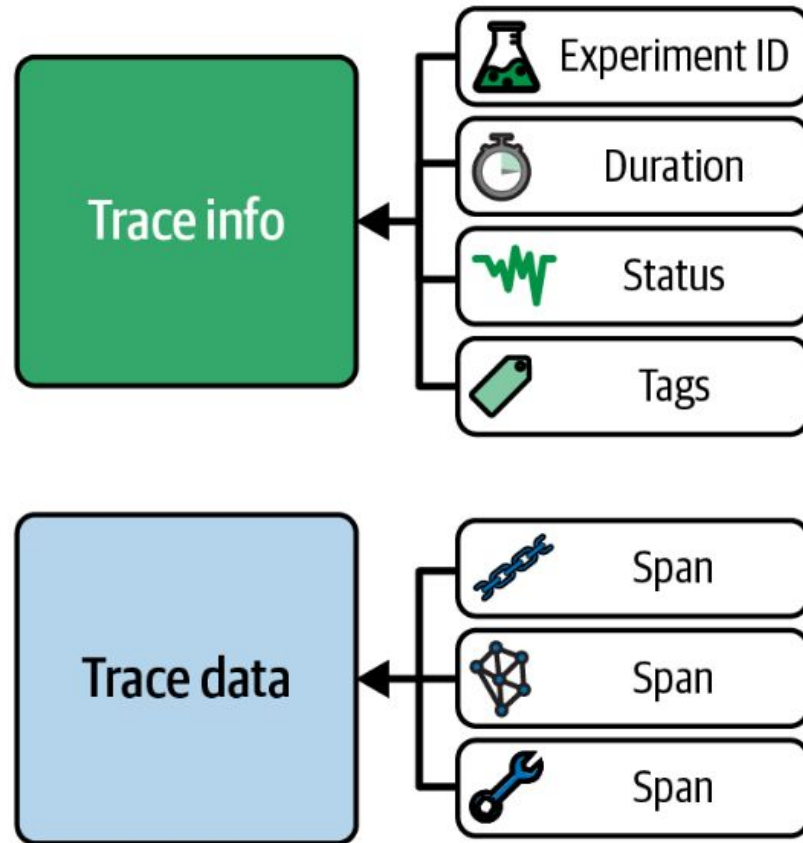


Figure 8-2. Trace info and trace data, adapted from [MLflow documentation](#)

LLM

This records language model inference.

TOOL

This records the execution of external tools, functions, and APIs.

RETRIEVER

This records the retrieval of relevant context, for example, from an AI search index.

MEMORY

This records instances of reading from or writing to conversational memory.

EMBEDDING

This records the generation of vector embeddings from text or other inputs.

RERANKER

This records the reranking of retrieved results based on relevance.

Los traces es nuestra manera para determinar la salud del sistema.

Sistema de Evaluación: Con criterio humano.

Explicit behavior

They force you to define exactly what “good behavior” looks like for your agent.

Measurable impact

They allow you to measure how changes to orchestration logic, prompts, or the underlying model affect performance.

System optimization

In many ways, choosing a prompt or a temperature setting is like tuning hyperparameters in classic ML; evals provide the data you need to make those choices.

Definir cómo queremos que se comporte nuestro sistema.

Medir campos en las medida por optimización de variables.

Sistema de Evaluación: Con criterio humano.

How to Evaluate LLM Systems

If you're starting with evals early, the best approach is a simple, iterative loop. Experts Aishwarya Naresh Reganti and Kiriti Badam, in their materials on AI Evals,¹ suggest this workflow:

1. Create examples: start small; 10–20 high-quality examples are better than 100 mediocre ones.
2. Run test cases: process them through your current agent.
3. Evaluate with the business: determine whether the output has met expectations.
4. Identify error patterns: explain exactly why an output failed.
5. Decide on key metrics: determine what you are actually measuring.
6. Iterate and expand: refine your prompts or tools and repeat.

Anthropic's [“Demystifying evals for AI agents” article](#) adds a few useful principles:

Product requirements naturally translate into test cases

If a requirement states that “*The agent must cite its sources*,” you can turn that into a test that verifies whether responses include references to the retrieved documents.

The “expert agreement” rule

A good test case is one in which two domain experts independently reach the same pass/fail verdict. If they disagree, the example is too noisy.

Test negative behaviors

Don't forget to test both cases in which a behavior should occur and where it shouldn't. Imbalanced datasets are just as problematic in agent evaluation as they are in traditional ML.

Hay distintas forma de medir estos sistemas, pero aquí se muestra un conjunto de pasos que han sido útiles para la comunidad Tech en el mundo.

Sistema de Evaluación: Métricas

Table 8-2. MLflow scorers²

Scorer	What it evaluates	Does it require expectations traces a session?
RelevanceToQuery	Does the response directly address the user's input?	No No No
Correctness	Are the expected facts supported by the response?	Yes No No
Completeness	Does the agent address all questions in a single user prompt?	No No No
Fluency	Is the response grammatically correct and naturally flowing?	No No No
ExpectationsGuidelines	Does the response meet specific expectations and guidelines?	Yes No No
Safety	Is the response free of harmful or toxic content?	No No No
Equivalence	Is the response equivalent to the expected output?	Yes No No

RetrievalGroundedness	Is the response grounded in retrieved information?	No Yes No
RetrievalRelevance	Are retrieved documents relevant to the user's request?	No Yes No
RetrievalSufficiency	Do retrieved documents contain all necessary information?	Yes Yes No
ToolCallCorrectness	Are the tool calls and arguments correct for the user query?	No Yes No
ToolCallEfficiency	Are the tool calls efficient without redundancy?	No Yes No
ConversationCompleteness	Does the agent address all user questions throughout the conversation?	No Yes Yes
ConversationalGuidelines	Do the assistant's responses comply with provided guidelines?	No Yes Yes
ConversationalRoleAdherence	Does the assistant maintain its assigned role throughout the conversation?	No Yes Yes
ConversationalSafety	Are the assistant's responses safe and free of harmful content?	No Yes Yes